

MS&E 342 Lecture 4:
Diffusion Factor Models: Generating
High-Dimensional Returns with Factor Structure

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Financial applications

- ▶ Enhance **stress testing and risk management** via different types of financial scenario generation
 - ▶ Multi-variate time series (Cont et al. '22; Seyfi et al. '22, Biagini et al. '24, Horvath et al. '24, Ni et al. '24)
 - ▶ Volatility surface (Vuletić and Cont '23, Cuchiero et al. '24, Jaimungal et al. '24)
 - ▶ Limit order books (Li et al. '20; Coletta et al. '23; Cont et al. '23, Reisinger et al. '24)
- ▶ **Assist decision making** with finance-focused LLM:
 - ▶ BloombergGPT (Wu et al. '23)
 - ▶ AlphaSense
- ▶ Better understand **climate risk and uncertainty** via climate scenario generation and forecasting
 - ▶ Rare climate event simulation (Peard and Hall '23)
 - ▶ Climate trajectory generation with different government policies

Challenges and remedies

- (1) While ML is often viewed as a “big data” tool, many foundational questions in finance are hindered by the “small data” reality
 - ▶ E.g, ~ 2000 samples of daily-returns in the past 10 years
- (2) Existing GenAI methods for financial applications are largely based on GANs, which suffer from instability, and lack of theoretical support...

Question: Can we develop a method that works in the data-scarce regime and enjoys a theoretical guarantee?

Our work

Question: Can we develop a method that works in the **data-scarce regime** and enjoys a **theoretical guarantee**?

When data exhibit **unknown low dimensional** factor structure, we design a structure-aware diffusion model (**Diffuson Factor Model**):

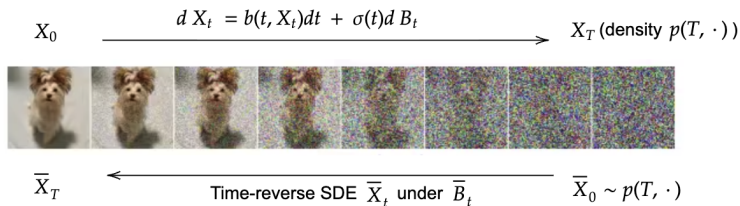
- (a) generate realistic high-dimensional asset returns
- (b) uncover latent factor subspace
- (c) enjoys efficient statistical guarantee: addresses the curse of dimensionality
- (d) applications in constructing mean-variance optimal portfolios and factor portfolios

Recap: Generative diffusion models

Consider the (forward) dynamics :

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \text{ with } X_0 \sim p_{\text{data}}(\cdot)$$

where $p_{\text{data}}(\cdot)$ is the data distribution (access to limited samples)



- Run the **time-reversed process** $\bar{X}$ starting at $\bar{X}_0 \sim p(T, \cdot)$, up to time T : $\bar{X}_T$ is a sample from $p_{\text{data}}(\cdot)$

Recap: Time-reverse process

Ingredient 1 (Anderson, '82; Haussmann and Pardoux, '86)

Let $p(t, \cdot)$ be the density of X_t , define

$$d\bar{X}_t = \bar{b}(t, \bar{X}_t)dt + \sigma(T-t)d\bar{B}_t, \text{ with } \bar{X}_0 \sim p(T, \cdot), \quad (\diamond)$$

where

$$\bar{b}(t, x) = -b(T-t, x) + \sigma^2(T-t) \underbrace{\nabla \log p(T-t, x)}_{\text{Stein's score function}}.$$

Then under mild conditions, $(\diamond)$ admits a strong solution and $\bar{X}_t$ has the same law as X_{T-t} .

Recap: time-reverse process

Recall that $\bar{\sigma}(t, x) = \sigma(T - t, x)$ and

$$\bar{b}(t, x) = -b(T - t, x) + \sigma^2(T - t) \nabla \log p(T - t, x)$$

Note that b, σ are known (part of the diffusion design). The “headache” is $p(t, \cdot)$, or $\nabla \log p(t, \cdot)$ (Stein’s score)!

Ingredient 2 (Explicit score matching)

Learn $\nabla \log p(t, \cdot)$ – score-matching (Song and Ermon, ’19; Ho, Jain, and Abbeel, ’20):

$$\min_{\theta} \mathbb{E} \left[\int_0^T \lambda(t) \|s_{\theta}(t, X_t) - \nabla \log p(t, X_t)\|_2^2 dt \right]$$

- ▶ $X_0 \sim p_{\text{data}}$, and $X_t|X_0$ follows the forward SDE we specify
- ▶ Needs good estimation of $\nabla \log p$ at each time t to recover p_{data}

Curse of dimensionality...

Question: How accurately can the score estimation be constructed with N samples from p_{data} in a d -dimensional space?

- ▶ [Chen et al. '23; Oko, Akiyama and Suzuki '23] established non-parametric analysis

$$\mathbb{E}_{t, X_t} \left[\|s_{\hat{\theta}}(X_t, t) - \nabla \log p(X_t, t)\|_2^2 \right] = \mathcal{O}(N^{-\frac{2}{d+2}})$$

- ▶ [Wibsono, Wu and Yang '24; Zhang et al. '24; Dou et al. '24] constructed a kernel density estimator $\hat{s}$ and achieved the rate

$$\mathbb{E}_x \left[\|s_{\hat{\theta}}(x) - \nabla \log p(x)\|_2^2 \right] = \mathcal{O}(N^{-\frac{2}{d+4}})$$

- ▶ Using gradient-descent algorithm, [Han, X. and Razaviyayn '24]

$$\mathbb{E}_{t, X_t} \left[\|s_{\hat{\theta}}(X_t, t) - \nabla \log p(X_t, t)\|_2^2 \right] = \mathcal{O}(N^{-\frac{2}{d+3}})$$

However, all these generic results suffer from the curse of dimensionality when d is big...

Factor model structure

In the context of finance, it is a common belief that there exists a **low-dimensional structure** underlying the high-dimensional asset returns...

Mathematically, assume $R \sim p_{\text{data}}$ has a **factor model structure** (with $k \ll d$)

$$R = \beta \underbrace{F}_{\substack{\text{unknown} \\ \text{factor}}} + \underbrace{\varepsilon}_{\substack{\text{idio.} \\ \text{residual}}}$$

- ▶ $R \in \mathbb{R}^d$: asset return vector on day i
- ▶ $F \in \mathbb{R}^k$ and $\varepsilon \in \mathbb{R}^d$ are independent rvs

Note that

- ▶ Most widely used model in the asset-pricing literature (e.g. CAPM, APT, Fama-French factor model, factor zoo...)
- ▶ Applications: covariance estimation, portfolio construction, strategy risk estimation, ...

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Why diffusion model is suitable for structured data?

Data structure leads to a **decomposition of $\nabla \log p_t$**

- ▶ Mathematically, consider the following simple forward process

$$dX_t = -\frac{1}{2}X_t dt + dB_t$$

with perfect linear data $X_0 = AZ$

- ▶ Take $A \in \mathbb{R}^{d \times k}$ with $A^\top A = I$, $Z \sim p_z$. Set $\alpha(t) = e^{-t/2}$ and $h(t) = 1 - e^{-t}$.
- ▶ Then we have the following result:

$$\nabla \log p(t, x) = \underbrace{A \nabla \log p_t^{LD}(A^\top x)}_{\text{On-support score in } \mathbb{R}^k} - \underbrace{\frac{1}{h(t)}(I - AA^\top)x}_{\text{Orthogonal score}}.$$

- ▶ On support score is in **low dimension** $\Leftarrow$ efficient to learn
- ▶ Orthogonal score is **linear** $\Leftarrow$ efficient to learn

Set-up.

The forward density is

$$p_t(x) = \int \phi_t(x \mid Az) p_z(z) dz, \quad \phi_t(x \mid Az) = \mathcal{N}(x; \alpha(t)Az, h(t)I).$$

Define $z' = A^\top x$ and

$$p_t^{LD}(z') = \int \phi_t(z' \mid z) p_z(z) dz,$$

where $\phi_t(z' \mid z) = \mathcal{N}(z'; \alpha(t)z, h(t)I)$.

Proof. Differentiate under the integral:

$$\nabla \log p_t(x) = \mathbb{E} \left[\frac{1}{h(t)} (\alpha(t) Az - x) \mid x \right].$$

Decompose $x = AA^\top x + (I - AA^\top)x$:

$$\nabla \log p_t(x) = \frac{1}{h(t)} \mathbb{E}[\alpha(t) Az - AA^\top x \mid x] - \frac{1}{h(t)} (I - AA^\top)x.$$

The first term lies in $\text{Im}(A)$:

$$\frac{1}{h(t)} \mathbb{E}[\alpha(t) Az - AA^\top x \mid x] = \frac{1}{h(t)} A (\alpha(t) \mathbb{E}[z \mid x] - A^\top x).$$

Since $p(z \mid x) = p(z \mid A^\top x)$, letting $z' = A^\top x$,

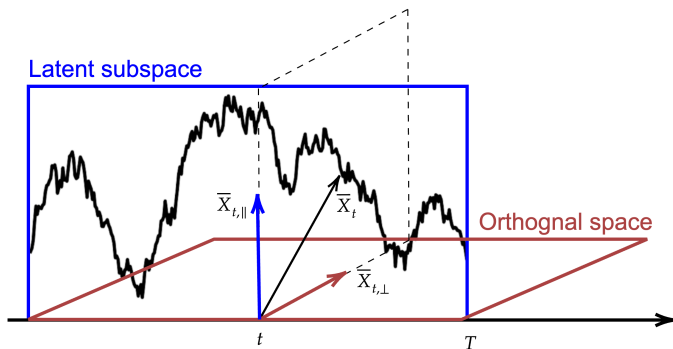
$$\nabla \log p_t^{LD}(z') = \frac{1}{h(t)} (\alpha(t) \mathbb{E}[z \mid z'] - z').$$

Substituting $z' = A^\top x$ gives the result. ■

Special case of perfect linear sub-space

Visual understanding

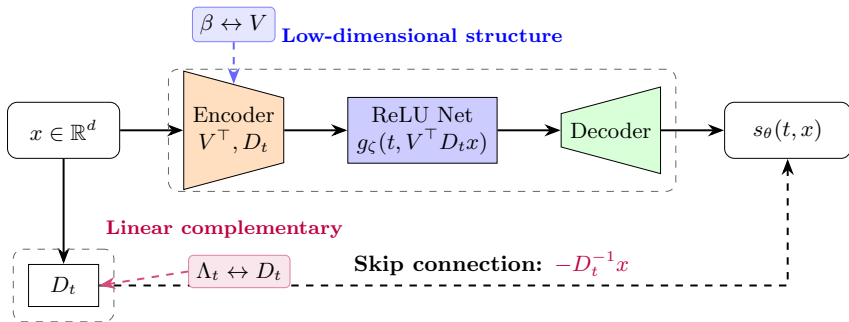
Intuition: on-support component carries **latent distribution information** and orthogonal component leads to the **subspace recovery**



Informed architecture design

Score decomposition $\Rightarrow$ Efficient score network architecture $\mathcal{S}_{\text{NN}}$:

$$\nabla \log p(t, x) = -\beta \left(\nabla \log p_t^{\text{fac}}(\Lambda_t \cdot \beta^\top \Lambda_t^{-1} x) - \beta^\top \Lambda_t^{-1} x \right) - \Lambda_t^{-1} x$$



- Does not require knowledge of the factors!

Theoretical guarantee: score function estimation

Theorem (Score estimation (Chen, X., Xu, Zhang '24))

If neural network $s_{\hat{\theta}}(t, x)$ minimizes the empirical risk loss with N samples,

$$\frac{1}{T - \delta} \int_{\delta}^T \|s_{\hat{\theta}}(t, \cdot) - \nabla \log p(t, \cdot)\|_{L^2(p_t)}^2 dt = \tilde{O} \left(\frac{1}{\delta} d^{\frac{5}{2}} N^{-\frac{2}{k+5}} \right).$$

- ▶ sample complexity depends primarily on k instead of d
- ▶ cannot remove the dependence on d completely due to the idio. noise
- ▶ δ represents early stopping in training
- ▶ **Other low-dim structures:** linear sub-space (Chen et al. '23), low-dim manifold (Zhang et al. '24), mixture Gaussian (Wang et al. '24), union of K subspaces (Chen et al. '24)
 - however, **perfect low-dim structure** is often too ideal to be true...

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Distribution guarantee and subspace recovery

Theorem (Distribution estimation (Chen, Xu, Zhang '25))

(1). By choosing $T = \frac{1}{k+5} \log N$ and $\delta = N^{-\frac{1}{k+5}}$, we have

$$\text{TV}(p_{\text{data}}, \hat{p}_{\delta}) = \tilde{\mathcal{O}}\left(d^{\frac{5}{4}} N^{-\frac{1}{2(k+5)}}\right),$$

where $\hat{p}_{\delta}$ is the distribution of the time-reverse SDE at $t = T - \delta$

(2). The corresponding k -dimensional eigenspace can be recovered using SVD on the sample covariance matrix with:

$$\|\hat{U}\hat{U}^{\top} - UU^{\top}\|_{\text{F}} = \tilde{\mathcal{O}}\left(N^{-\frac{1}{k+5}}\right)$$



► **Key Advantage:** do not need prior knowledge on the factors F and the subspace can be recovered (subject to small error)

Synthetic experiment: covariance estimation

- ▶ $k = 16$ factors, $d = 2048$ assets
- ▶ factors F and residual noise ε are Gaussian (Bai and Ng '02, '23).

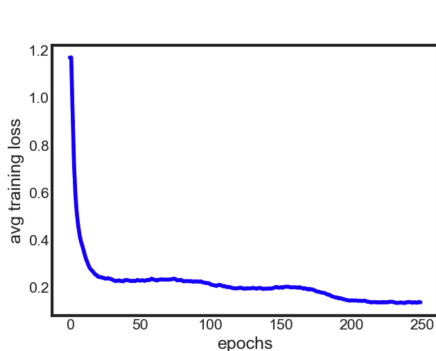


Figure 1: Average training loss

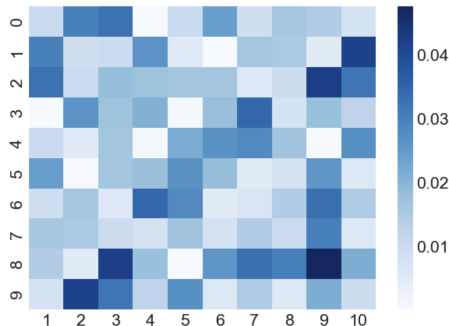


Figure 2: Relative error between sample covariance matrix and ground-truth covariance matrix

Synthetic experiment: latent subspace recovery

- ▶ Direct estimation: PCA on samples (N)
- ▶ Diffusion model: PCA on generated data from diffusion factor model

Sample (N)	Diffusion model RE	Direct estimation RE	Error ratio
$2^9 = 512$	0.247 (± 0.012)	0.284	0.869 (± 0.044)
$2^{10} = 1024$	0.202 (± 0.006)	0.218	0.926 (± 0.028)
$2^{11} = 2048$	0.153 (± 0.005)	0.159	0.960 (± 0.033)
$2^{12} = 4096$	0.110 (± 0.004)	0.109	1.009 (± 0.036)
$2^{13} = 8192$	0.085 (± 0.004)	0.084	1.012 (± 0.047)

Table 1: Relative error (RE) of recovered latent subspace (RE

$$:= \|\hat{U}\hat{U}^\top - UU^\top\|_F / \|UU^\top\|_F)$$

- ▶ **Phase transition phenomena:** diffusion model has better performance when $N \leq 2^{11}$; otherwise comparable with direct estimation
- ▶ $2^{11} \approx 2000$: 10 years of daily data

Takeaway: diffusion model well captures structural information and performs well in the small data regime

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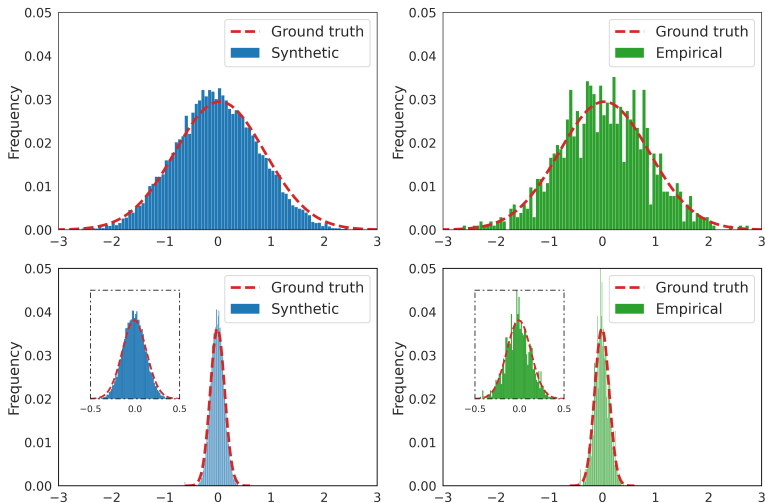
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Takeaway: diffusion model well captures **structural information** and performs well in the **small data regime**

Why diffusion model is powerful in the small data regime?



*Figure 3: Stock return data simulated from **diffusion model** (left) and **ground-truth** (right), for the assets with maximum (up) and minimum variance (down)*

Empirical analysis: mean-variance portfolio

Daily return data for U.S. stocks from 2001/05/01 to 2024/04/30.

Mean-variance optimization:

$$\max_{\omega} \omega^{\top} \mu - \frac{\eta}{2} \omega^{\top} \Sigma \omega, \quad \text{subject to } \omega^{\top} \mathbf{1} = 1 \text{ and } \|\omega\|_{\infty} \leq 0.05 \quad (\star)$$

Classical Methods (only observed data)

1. EW Method: equal-weighted portfolio
2. VW Method: value-weighted (using market cap) portfolio
3. Emp+Emp Method: sample mean and sample covariance to solve
($\star$)
4. Emp+LW Method: sample mean and shrinkage covariance to solve
($\star$)

Diffusion-Based Methods (diffusion-generated data)

5. Diff+Emp+Emp Method: Emp+Emp Method + generated data
6. Diff+Emp+LW Method: Emp+LW Method + generated data
7. Real_Emp+Diff_Emp Method: Sample mean using observed data and sample covariance matrix using diffusion-generated data

Empirical analysis: mean-variance portfolio

Method	Mean	SR	CER
Diff+Emp+Emp	0.15	0.80	0.10
Diff+Emp+LW	0.12	0.72	0.08
Real_Emp +Diff_Emp	0.09	0.61	0.06
EW	0.10	0.43	0.02
VW	0.09	0.41	0.02
Emp+Emp	0.01	0.07	-0.02
Emp+LW	0.02	0.15	-0.01

Table 2: Portfolio performance with 20 bps transaction cost for $\eta = 3$ (SR: Sharpe ratio, CER: objective value)

- ▶ Diff+Emp outperforms all alternatives, including EW, Shr, Diff+Shr
- ▶ Improvement in both mean and covariance estimation
- ▶ Shrinkage techniques no longer needed with diffusion-generated data

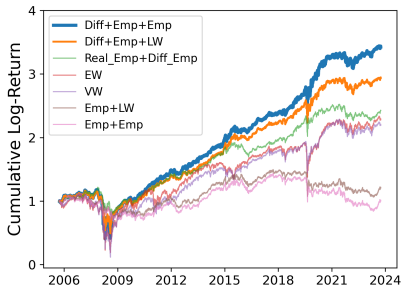


Figure 4: Out-of-sample cumulative log-returns (NASDAQ ITCH)

Empirical analysis: factor portfolio

Factor tangency portfolio:

$$\max_{\omega} \frac{\omega^{\top} \mu_{\text{fac}}}{\sqrt{\omega^{\top} \Sigma_{\text{fac}} \omega}}, \quad \text{subject to } \omega^{\top} \mathbf{1} = 1$$

Classical Methods (only observed data)

1. FF Method: Fama-French 5 factors (FF5), momentum (MOM), reversal (ST-Rev, LT-Rev) (Carhart '97, Fama and French '15)
2. PCA Method: Principal Component Analysis (PCA)—extract factors
3. POET Method: Principal Orthogonal ComplEment Thresholding (POET) (Fan et al. '13)—extract factors.
4. RPPCA Method: Risk-premia PCA (RP-PCA) (Lettau and Pelger '20)

Diffusion-Based Methods (diffusion-generated data)

5. Diff+PCA Method: Apply PCA on diffusion-generated data
6. Diff+POET Method: Apply POET on diffusion-generated data
7. Diff+RPPCA Method: Apply RP-PCA on diffusion-generated data

Empirical analysis: factor portfolio

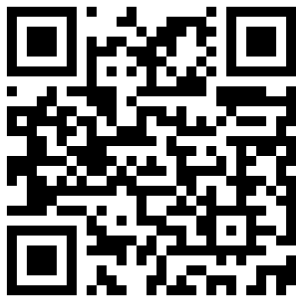
# Factor	3	5	6	8	Method	1	2	3
Diff+PCA	1.805	2.158	2.322	2.631	Diff+PCA	Mkt-RF	LT-Rev	MOM
Diff+POET	1.841	2.178	2.339	2.739		(0.725)	(0.404)	(0.395)
Diff+RPPCA	1.985	2.367	2.550	2.810	Diff+POET	Mkt-RF	LT-Rev	MOM
FF	0.648	0.726	0.861	0.881		(0.758)	(0.434)	(0.419)
PCA	0.402	0.453	0.528	0.673	Diff+RPPCA	Mkt-RF	LT-Rev	MOM
POET	0.872	0.930	1.356	1.463		(0.710)	(0.454)	(0.382)
RPPCA	0.631	1.250	1.701	1.892				

Table 3: Out-of-sample Sharpe ratios of factor tangency portfolios

Table 4: Most correlated factors and absolute correlations between Diff methods and FF method (top 3)

- ▶ Capture systematic risk with higher factor risk premia
- ▶ Capture interpretable economic characteristics

Thank you!



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